

VoxCAD: A 0.82-to-81.0mW Intelligent 3D-Perception dToF SoC with Sector-Wise Voxelization and High-Density Tri-Mode eDRAM CIM Macro

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Abstract:
VoxCAD, an intelligent low-power dToF SoC for end-to-end 3D perception applications, is presented with three features: 1) LiDAR dToF sensing-integrated 2D-ROI guided poin... [Show More](#)

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 Contents

The rapid growth of the electric-vehicle (EV) and smart robotics market drives demand for full self-driving (FSD), advanced driver-assistance systems (ADAS) and autonomous navigation. The foundation of such systems lies in end-to-end environment perception from 2D/3D sensors and efficient execution of AI tasks. [Figure 7.2.1](#) presents a conventional 3D perception flow, beginning with a range sensor for retrieving object distances, followed by the transferring of the sensor data with user datagram protocol (UDP)/serial interfaces (I/Fs) to a preprocessor for point-cloud construction. The voxelized point clouds are then stored in external memory, and finally processed by CPUs/GPUs for AI tasks such as 3D object detection, segmentation, and classification. Yet, CPUs/GPUs lack application-specific optimizations resulting in low energy efficiency and high latency.

Recently, several application-specific digital processors have emerged to improve energy efficiency and latency on 3D perception tasks [\[1\]– \[3\]](#), but still suffer from three critical challenges. First, the sensor data transfer via UDP/serial I/Fs dominates performance, using 60 to 72% of system energy and accounting for 51 to 65% of latency [\[1\]](#). Even though on-chip sensor integration in [\[4\]– \[6\]](#) reduces transfer costs, this approach can only demonstrate 2D imaging and simple AI tasks (e.g., face recognition). Also, the implemented singlephoton avalanche diode (SPAD)-based vision SoC with 2D/3D imaging capability in [\[7\]](#) still suffers from a limited energy efficiency and latency due to the fully digital implementation. Furthermore, despite the improved energy efficiency and latency offered by 3D stacking [\[6\], \[8\]](#), such techniques can still incur a high fabrication and packaging cost. Second, voxelization workloads require massive external memory access (EMA), which can degrade system performance by up to $7\times$ in [\[1\]](#). To reduce EMA, [\[2\]](#) stores voxels fully on-chip, but inevitably demands an impractically large on-chip memory (5.9 Mbit). Third,



although mainstream 3D transformers allow higher accuracy, they can introduce an excessive hardware cost due to the $2\times$ increase in parameters [9], [10], not to mention the underutilization and low energy efficiency of CNN-specific processors due to different data reusability on mixed CNN/attention operations [1]– [3].

To resolve the above challenges, this work presents an ultra-low power intelligent SoC, called VoxCAD, with key features including: 1) sensing-integrated 2D region-of-interest (ROI)-guided point cloud construction for reducing sensor data transfer latency, energy, and EMAs; 2) sector-wise voxelization with a central-computation unit (CCU), enabling full on-chip voxel management with lower memory requirements and EMA reduction; and 3) reconfigurable tri-mode eDRAM compute-in-memory (CIM) macros based on hybrid 2T1C and 1T1C cells to maximize the storage density and area efficiency, while supporting both 2D CNN and attention workloads.

[Figure 7.2.2](#) depicts the SoC with a 128×1 line SPAD direct time-of-flight (dToF) sensor. It supports 256×128 2D intensity and 512×128 3D distance output via scanning operation with galvo mirror. The dToF operates at 600 MHz with an on-chip PLL, or 200 MHz with an external clock to support always-on tasks. The tightly coupled preprocessor receives distance data with ultra-low latency and energy, and then performs point cloud construction and radius outlier removal (ROR) denoising. This work employs four CCUs for point cloud voxelization and feature extraction. Each CCU includes a point-to-voxel unit, a programmable SIMD, a run-length encoder (RLE), and a look-up table (LUT). The point-to-voxel units support point cloud voxelization through quantization, rounding, and translation lookaside buffer (TLB)-based address translation. The eight reconfigurable eDRAM CIM macros (516 Kbit total) support high-density storage and energy efficient computation. A RISC host core with two global 8KB SRAMs handle system control and data buffering, while a 64b 2D mesh network-on-chip (NoC) interconnects all processing modules.

[Figure 7.2.3](#) presents the 2D/3D dToF sensing-integrated preprocessor and 2D-ROI guided point cloud construction flow. Each dToF pixel integrates a SPAD, an analog front-end, and an up/down counter. In 2D mode, the sensor delivers 128×256 8b intensity maps based on photon counting at 20 frame per second (fps) with on-chip PLL, or at 1 fps in ultra-low-power mode. In 3D mode, the up/down counting scheme with coarse-fine coding resolves the depth information [11], yielding a 9b distance resolution. The preprocessor streams up to 1152 b /cycle of intensity/distance data into eDRAM macros, where the lightweight 2D CNN identifies regions of interest (ROI) such as cars, pedestrians, obstacles, and other objects. We divide the intensity data into multiple azimuth sectors based on rotation angles. Only the ROI-flagged azimuth sectors require distance measurement, while the unflagged ones are skipped for reducing point cloud scale. We construct the measured distance into point clouds with ROR denoising. Measurement results demonstrate that this sensing-integrated ROI-guided flow reduces transfer energy and latency by more than 99% compared to UDP solutions, and suppresses 82% of raw points with only a 2.8% mean average precision (mAP) loss. It further achieves a 70 and 77% reduction in the system's end-to-end inference latency and energy consumption, respectively.

[Figure 7.2.4](#) illustrates the CCU with sector-wise voxelization flow. To reduce the on-chip storage requirement, VoxCAD employs a coarse-to-fine sector division strategy. Initially, we partition the point clouds into 'M' coarse azimuth sectors, with each further subdivided into 'N' fine sectors along distance bins. However, the non-uniform point cloud density across near and far distances can still compromise the on-chip memory requirement. This work employs logarithmic distance bins to balance memory allocation among the near-field dense points and far-field sparse points. Each fine sector is processed sequentially in a streaming manner to reduce the on-chip memory size. Specifically, first we quantize and round the points to voxel indices, and forward the results to the TLB. TLB previews the sector, and constructs a voxel table to record point counts per voxel. Guided by the voxel table, the RISC core allocates contiguous SRAM blocks for non-empty voxels, which are then processed by the SIMD to obtain both the mean position and cluster features. Finally, we transfer the voxel-level features to the eDRAM CIM macros for higher-dimensional feature extraction. Measurement results show that this sector-wise voxelization flow eliminates the need for a large on-chip buffer, reduces EMA of activation by nearly 99%, and achieves an overall EMA reduction of up to 87% at a division level of 'M = 8' and 'N = 6' with only 1% mAP loss. This offers a trade-off between sector granularity and model parameter EMA overhead.

[Figure 7.2.5](#) shows the reconfigurable tri-mode eDRAM macros, integrating buffers, 256×258 1T1C storage cells, 16×128 2T1C compute cells, a word-line (WL) decoder, WL drivers, bit-line (BL) drivers, local sense

amplifiers (LSAs), global sense amplifiers (GSAs), and 4 SAR-ADCs. Each macro comprises 16×128 local cell arrays (LCAs), with each containing 16×2 1T1C cells, one 2T1C cell, and one LSA. It supports three operation modes: 1) CIM mode for matrix multiplication; 2) memory mode for tensor storing; and 3) XNOR mode for similarity checking. In CIM mode, the index buffer stores the non-zero value index, with the all-zero value operations skipped for achieving high area and energy efficiency. Non-zero weights are stored in 1T1C cells, then loaded to the local bit-line (LBL) and further amplified by LSA. Non-zero activation controls the gate of 2T1C cells to charge or discharge the coupled capacitor. Afterwards, the LCAs perform intra-inter charge sharing (for single-bit activation, multi-bit weight, and multi-channel addition), and the SAR-ADCs convert the final voltage into digital values. CCU supports shift-add operations for multi-bit accumulation. The simultaneous operation of multiple LCAs contributes to high utilization for high area efficiency. For memory mode, we write the data to 1T1C cells via global bit-line (GBL), and readout by LSA and GSA. We first configure half macros to memory for storing Q/K/V tensors, and later reconfigure them to CIM to reduce data movement on attention layers. In XNOR mode, weight and activation are represented in a complementary format. The gate of 2T1C is always on, and we load the activation on WL & WLB to readout the stored complementary weight in capacitors. The bitwise XNOR operation is accomplished by capacitor charging. LCAs then perform charge sharing to obtain multi-tensor similarity and readout by ADCs. The system initializes with low power always-on mode, and it performs similarity checking between 2 adjacent intensity frames at 1 fps. If the similarity does not exceed the threshold, the system remains in low-power mode to conserve energy, and performs 2D-ROI detection and 3D detection otherwise. By using high-density 1T1C cells for storing and 2T1C cells for computing, the macro achieves a $9.2\times$ storage density increase compared to 2T1C-only macro [12]. Further, the intra/inter charge sharing with sparsity exploration contributes to a $34.2\times$ area efficiency improvement compared to previous works [12], [13].

Figure 7.2.7 illustrates the 130 nm chip photo, summary table, and power measurement setup. The processing modules operate at a supply voltage from 1.1 to 1.7 V, with an operating frequency from 10 to 250 MHz. The dToF operates at 600 MHz with an on-chip PLL, and switches to ultra-low power always-on mode with a 200 MHz external clock, based on temporal similarity checking between two adjacent intensity frames at 1fps with CIM macros. Figure 7.2.6 presents the measured SoC detection results and benchmarks this work with the state of the art. Using the Kitti dataset and our measured point clouds, this work achieves 80.3 and 75.6 mAP on the dynamic sparse voxel transformer (DSVT) [10] backbone, respectively. With a 97.5% reduction in system energy and a 91.2% reduction in latency, this work also demonstrates a 101-to-406x density FoM improvement compared to digital point cloud processors [1]– [3], and a 12.7-to-153x improvement over prior DRAM macros [12], [13] through the integration of high-density eDRAM macros. It also exhibits an end-to-end energy efficiency improvement of 3.51 to 6.85x, and a latency reduction of 2.4 to 22.9x compared to digital point cloud processors [1]– [3] through the co-optimization of sensor and processor. Compared to previous sensor-integrated smart vision SoCs [4]– [7], it offers an energy efficiency improvement of 9.5 to 60.5x and a latency reduction of 2.1 to 26.1x using the eDRAM CIM macro with sparsity exploration.



Figure 7.2.1:
Design challenges of end-to-end 3D perception systems.



Figure 7.2.2:
Intelligent 3D perception dToF SoC overview with three key features.



Figure 7.2.3:
SPAD dToF sensing-integrated 2D-ROI guided point cloud construction flow.



Figure 7.2.4:
Sector-wise voxelization flow with central-computation unit based on coarse-fine sector division.



Figure 7.2.5:
Tri-mode eDRAM CIM macros with hybrid 1T1C/2T1C cells for 3D transformer and similarity checking.



Figure 7.2.6:
Measurement results and performance comparison table.



Figure 7.2.7:
Chip photo, performance summary and system power measurement setup.

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